

Introduction to Tokens and Tokenization

Spoken Tutorial

brought to you by

**EduPyramids Educational Services
Pvt. Ltd.**



<https://spoken-tutorial.org>

Learning Objectives

In this tutorial, we will learn,

- ▶ **Explain what tokens are, how tokenization works, and why tokens are crucial for LLM performance, cost, and efficiency.**



System Requirements

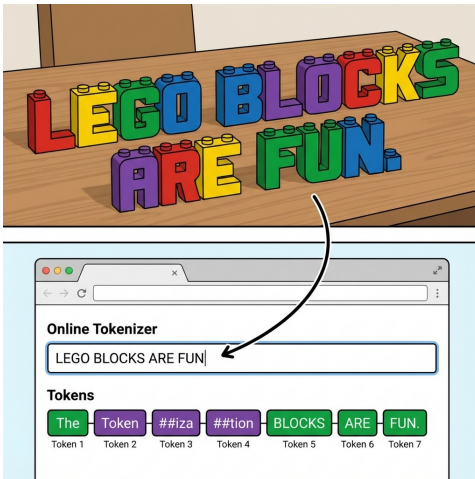


Pre-requisite Slide

- ▶ **Introduction to Generative AI,
Introduction to Prompting**



What are Tokens?

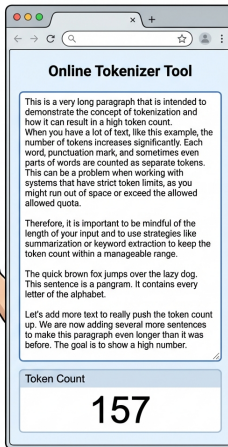
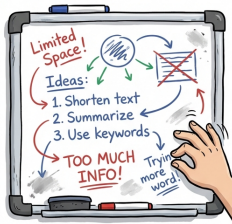


How Tokenization Works

- ▶ The tokenizer acts like a special dictionary for the LLM.
- ▶ It knows how to split raw text into these sub-word tokens



Why Tokens Matter: Context Windows

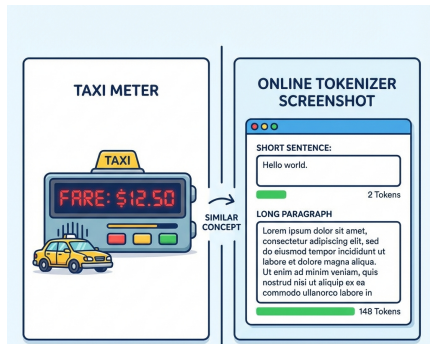


Why Tokens Matter: Cost

- ▶ Second, tokens directly impact the cost of using LLMs.
- ▶ API providers charge you per token processed.



It's like a taxi



Why Tokens Matter: Chunking

- ▶ Finally, tokens are vital for chunking your documents.
- ▶ When preparing large documents for an LLM, we often break them into smaller



Summary

In this tutorial, we learned that:

- ▶ **With this, we come to the end of this tutorial.**
- ▶ **Tokenization is the process of splitting text into these sub-word units.**
- ▶ **And tokens are critical for context windows, cost, and efficient chunking.**



Assignment

- ▶ Find an online tokenizer tool.
- ▶ Experiment with different types of text: short words, long words, sentences,...
- ▶ Observe how the token count changes.
- ▶ Try to find a sentence that results in exactly 10 tokens.



Thank you

